First overview of governing equations

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1 A note on notations

1.1 Material derivative

The following equations are all equivalent, which can be shown algebraically.

$$\begin{aligned}\frac{d}{dt} &= \frac{\partial}{\partial t} + u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z} \\ &= \frac{\partial}{\partial t} + \mathbf{u}\cdot\nabla \\ &= \frac{\partial}{\partial t} + \mathbf{u}\cdot\operatorname{div}()\end{aligned}$$

1.2 Tensors

$$\tau_{ij}\frac{\partial u_i}{\partial x_j} = \boldsymbol{\tau} : \nabla\mathbf{u}$$

where

$$\nabla\mathbf{u} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix}$$

The : is called the double contraction operator or the matrix inner product. It contracts two tensors into a scalar. Since τ is the symmetric stress tensor,

2 Starting equations (partial differential equations)

2.1 Energy equation

2.1.1 CRB

CRB: Cushman-Roisin and Beckers, 2011

$$\rho C_v \frac{dT}{dt} + p\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right) = k_T \nabla^2 T \tag{1}$$

I refer to Schubert et al., 2001 for the conversion into the constant-pressure heat capacity variant of this equation.

2.1.2 Jarvis & McKenzie, 1980

jamc80 Jarvis and Mckenzie, 1980:

$$\rho C_p\left(\frac{\partial T}{\partial t} - \frac{\alpha T}{\rho C_p}\frac{\partial P}{\partial t} + \mathbf{u}\cdot[\nabla T - (\nabla T)_s]\right) = \nabla\cdot(K\nabla T) + H + \tau_{ij}\frac{\partial u_i}{\partial x_j} \tag{2}$$

2.1.3 Gassmöller et al., 2020

Gassmöller: Gassmöller et al., 2020

$$\rho C_p \left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) - \nabla \cdot (k \nabla T) = \rho H + \tau : \dot{\varepsilon} + \alpha T (\mathbf{u} \cdot \nabla p) + Q_L \tag{3}$$

2.1.4 Schubert et al., 2001

Schubert et al., 2001 presents three thermmal energy equations:

The general form of the energy equation (6.9.1) is:

$$\rho T \frac{Ds}{Dt} = \tau_{ij} \frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) + \rho H \tag{4}$$

For an incompressible Newtonian fluid (eq. 6.9.13 + 6.9.14):

$$\rho C_v \frac{DT}{Dt} = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \tag{5}$$

with $\frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2$ the viscous dissipation function.

In terms of C_p instead of C_v (for a compressible fluid) (eq. 6.9.16):

$$\rho C_p \left(\frac{DT}{Dt} - (\frac{\partial T}{\partial p})_s \frac{Dp}{Dt} \right) = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \tag{6}$$

with (eq. 6.9.15, 6.9.17):

$$(\frac{\partial T}{\partial p})_s = \frac{\alpha T}{\rho C_p}$$
$$(\frac{DT}{Dt})_s = (\frac{\partial T}{\partial p})_s \frac{Dp}{Dt} = \frac{\alpha T}{\rho C_p} \frac{Dp}{Dt}$$

2.2 Momentum equation

2.2.1 CRB

CRB: Cushman-Roisin and Beckers, 2011 per component

$$x : \rho \left(\frac{du}{dt} + f_* w - f v \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau^{xx}}{\partial x} + \frac{\partial \tau^{xy}}{\partial y} + \frac{\partial \tau^{xz}}{\partial z} \tag{7a}$$

$$y : \rho \left(\frac{dv}{dt} + f u \right) = -\frac{\partial p}{\partial y} + \frac{\tau^{xy}}{\partial x} + \frac{\partial \tau^{yy}}{\partial y} + \frac{\partial \tau^{yz}}{\partial z} \tag{7b}$$

$$z : \rho \left(\frac{dw}{dt} - f_* u \right) = -\frac{\partial p}{\partial z} - \rho g + \frac{\tau^{xz}}{\partial x} + \frac{\tau^{yz}}{\partial y} + \frac{\partial \tau^{zz}}{\partial z} \tag{7c}$$

2.2.2 Jarvis & McKenzie, 1980

jamc80, eq. 4 Jarvis and Mckenzie, 1980:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla P + \rho \mathbf{g} + \frac{\partial}{\partial x_j} \tau_{ij} \tag{8}$$

2.2.3 Gassmöller et al., 2020

Gassmöller Gassmöller et al., 2020:

$$-\nabla \cdot \tau + \nabla p = \rho \mathbf{g} \tag{9}$$

2.2.4 Schubert et al. 2001

$$\rho \frac{\mathrm{d}u_i}{\mathrm{d}t} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i \tag{10}$$

In 2D, the momementum equation components are:

$$\rho \left(\frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y}, \tag{11}$$

$$\rho \left(\frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \rho g. \tag{12}$$

Note that the assumption is made that gravity acts in the positive y-direction. Section 6.14?

2.3 Mass equation (continuity equation) (trivial)

jamc80, eq. 3 Jarvis and Mckenzie, 1980:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \tag{13}$$

CRB Cushman-Roisin and Beckers, 2011:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 \tag{14}$$

Gassmöller Gassmöller et al., 2020:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \tag{15}$$

schubert et al. (2001) Schubert et al., 2001

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_1}(\rho u_i) = \frac{1}{\rho}\frac{D\rho}{Dt} + \frac{\partial u_1}{\partial x_1} = 0 \tag{16}$$

where $\frac{D}{Dt}$ represents the material derivative (**??**). For incompressible flow, $\frac{D\rho}{Dt} = 0$, so eq 16 reduces to:

$$\frac{\partial u_i}{\partial x_i} = \nabla \cdot \mathbf{u} = 0 \tag{17}$$

2.4 Starting equations (constitutive equations)

2.5 Initial and boundary conditions

3 Appendix 1: tables of constants and reference/scaling values used

4 Schubert et al., 2001 General equations

This document/section has a different approach to structure the derivation of the ALA: this is a summary of the derivation of the ALA done by Schubert et al., 2001. red text means assumtions/approximations were made toward the ALA.

Table 1: Constants used by Schubert et al., 2001

Constant	Value	Name
k_b	0	Bulk viscosity
α		thermal expansivity
χ_T		isothermal compressibility
c_v		specific heat at constant volume
c_p		specific heat at constant pressure
χ_a		adiabatic compressibility
k		thermal conductivity
H		rate of internal heat production per unit mass
K_a		adiabatic bulk modulus
K_T		isothermal bulk modulus

Table 2: Reference values used by Schubert et al., 2001

Symbol	Value	Description
ΔT		Characteristic temperature difference driving thermal convection
χ_{T_r}		Representative isothermal compressibility
α_r		Representative thermal expansivity
u_r	$\frac{k_r}{\rho_r c_{p_r} b}$	Velocity scale
μ_r	10^{21} Pa s	Representative viscosity
χ_a		Adiabatic compressibility
b		Depth of convecting region
p_r	$\frac{\mu_r u_r}{b}$	Scale for dynamic pressure variations
k_r	4 W m^{-1} K^{-1}	Reference value thermal conductivity
c_{p_r}	1 kJ kg^{-1} K^{-1}	Reference specific heat at constant pressure

4.1 Mass equation/continuity equation/conservation of mass

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i) = \frac{1}{\rho}\frac{D\rho}{Dt} + \frac{\partial u_i}{\partial x_i} = 0 \tag{18}$$

where $\frac{D}{Dt}$ represents the material derivative ((1.1)). For incompressible flow, $\frac{D\rho}{Dt} = 0$, so Eq. (18) reduces to:

$$\frac{\partial u_i}{\partial x_i} = \nabla \cdot \mathbf{u} = 0 \tag{19}$$

For the special case of an incompressible fluid, $\frac{D\rho}{Dt} = 0$, so Eq. (19) reduces to

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{20}$$

Table 3: Dimensionless parameters and their values for the Earth's mantle

Parameter	Definition	Value	Description
Pr	$\frac{\mu_r c_{p_r}}{k_r}$	2.5×10^{23}	Prandtl number
M^2	$\frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2}$	10^{-33}	Mach number (squared)
ϵ	$\alpha_r \Delta T_r$		Fractional density change due to temperature variations

4.2 Momentum equations/conservation of momentum

Starting governing equation (6.4.1):

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i \tag{21}$$

we assume gravity g to be the only body force.The coriolis force is negligibly small in the Earth's mantle, and the centrifugal force due to the Earth's rotation is included in the gravity term.

In 2D, the momementum equation components are:

$$\rho \left(\frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y}, \tag{22}$$

$$\rho \left(\frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \rho g. \tag{23}$$

Note that the assumption is made that gravity acts in the positive y-direction.

4.3 Navier-Stokes equation

The local strain rate tensor e_{ij} is given by:

$$e_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \tag{24}$$

For a Newtonian, isotropic fluid, the relation between the deviatoric(!) stress tensor and the strain rate tensor is

$$\tau_{ij} = 2\mu e_{ij} + \lambda e_{kk} \delta_{ij} \tag{25}$$

Defining the bulk viscosity k_B:

$$\frac{\tau_{ii}}{3} = e_{ii} \left(\lambda + \frac{2}{3}\mu \right) = k_B e_{ii} \tag{26}$$

The combination of Eq. (25), (26) and (24) gives

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \left(k_B - \frac{2}{3}\mu \right) \frac{\partial u_k}{\partial x_k} \delta_{ij} \tag{27}$$

and substitution of Eq. (27) into Eq. 21 gives

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \left(k_B - \frac{2}{3}\mu \right) \frac{\partial u_k}{\partial x_k} \delta_{ij} \right] + \rho g_i \tag{28}$$

For many fluids, k_B is taken to be zero **(Stokes assumption)**, so Eq. (27) becomes

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right) \tag{29}$$

and Eq. (28) becomes the Navier-Stokes equation:

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right) \right] + \rho g_i \tag{30}$$

for incompressible flow, Eq. (19) tells us $\frac{\partial u_k}{\partial x_k} = 0$, and together with a constant dynamic viscosity μ so Eq. (30) reduces to:

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \mu \frac{\partial^2 u_i}{\partial x_j^2} + \rho g_i \tag{31}$$

This equation represents a balance between inertial forces $\left(\frac{Du_i}{Dt}\right)$, pressure forces $\left(\frac{\partial p}{\partial x_i}\right)$, viscous forces $\left(\mu \frac{\partial^2 u_i}{\partial x_j^2}\right)$ and the body force due to gravity (ρg_i)

4.4 Thermal energy equations/conservation of energy

Schubert et al., 2001 presents various different formulations of the energy equation.

The general form of the energy equation (6.9.1) is:

$$\rho T \frac{Ds}{Dt} = \tau_{ij} \frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) + \rho H \tag{32}$$

The terms on the right side of Eq. (32) are, respectively, volumetric heat production rates due to viscous dissipation (internal friction), thermal conduction, and internal heat generation (radiogenic?). Where it is assumed that Fourier's law of heat conduction for an isotropic medium is applicable:

$$q_i = -k \frac{\partial T}{\partial x_i} \tag{33}$$

Where q_i are the components of the heat flux vector. Additionally, the viscous dissipation term can be written as the dissipation function:

$$\Phi = \tau_{ij} \frac{\partial u_i}{\partial x_j} \tag{34}$$

Inserting the definition of our deviatoric stress tensor with $k_B = 0$ (Eq (29)), this becomes

$$\Phi = 2\mu \left\{ \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{1}{3} \left(\frac{\partial u_k}{\partial x_k} \right) \delta_{ij} \right\}^2 \tag{35}$$

For an incompressible and Newtonian fluid ($\frac{\partial u_i}{\partial x_i} = 0$), the dissipation function reduces to

$$\Phi = \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 \tag{36}$$

With various thermodynamic relationships:

$$\alpha = \frac{1}{v} \left(\frac{\partial v}{\partial T} \right)_p \tag{37}$$

$$c_p = T \left(\frac{\partial s}{\partial T} \right)_p \tag{38}$$

$$\left(\frac{\partial s}{\partial p} \right)_T = - \left(\frac{\partial v}{\partial T} \right)_p \tag{39}$$

It can be shown that

$$\frac{Ds}{Dt} = \frac{c_p}{T}\frac{DT}{Dt} - \frac{\alpha}{\rho}\frac{Dp}{Dt} \tag{40}$$

Plugging 40 into Eq. (32) gives

$$\begin{aligned} \rho c_p \frac{DT}{Dt} - \alpha T \frac{DP}{Dt} &= \frac{\partial}{\partial x_i}\left(k\frac{\partial T}{\partial x_i}\right) + \Phi + \rho H \\ \rho c_p \left(\frac{DT}{Dt} - \frac{\alpha T}{\rho c_p}\frac{DP}{Dt}\right) &= \frac{\partial}{\partial x_i}\left(k\frac{\partial T}{\partial x_i}\right) + \Phi + \rho H \end{aligned} \tag{41}$$

5 Schubert et al., 2001 Approximate equations

In this section, Schubert derives the approximate forms of the equations of conservation of energy and momentum appropriate to mantle convection.

5.1 Linearizing the equation of state (density ρ)

From now on, density ρ is considered a function of temperature T and pressure p only, neglecting the density changes associated with mineral phase transitions and changes in chemical composition.
Most of the density change in the mantle is due to hydrostatic compression. The mantle density increases by approximately 65% between the surface and the core-mantle boundary. This is Schubert's justification for assuming a linearized equation of state for mantle density:

$$\begin{aligned} \rho &= \overline{\rho}(\overline{T},\overline{p}) + \rho' \\ \rho &= \overline{\rho}(\overline{T},\overline{p}) + \left(\overline{\frac{\partial\rho}{\partial p}}\right)_T p' + \left(\overline{\frac{\partial\rho}{\partial T}}\right)_p T' \\ \rho &= \overline{\rho}(\overline{T},\overline{p}) + \overline{\rho\chi_T}p' - \overline{\alpha\rho}T' \end{aligned} \tag{42}$$

with

$$\begin{aligned} T &= \overline{T} + T' \\ p &= \overline{p} + p' \end{aligned} \tag{43}$$

Where the overbar ($\overline{\rho}$) refers to a reference state and the primes (ρ') departures from it. The reference state is assumed motionless and steady:

- motionless: $\overline{u_i} = 0$
- steady: $\frac{\partial\overline{u_i}}{\partial t} = 0$
- motionless and steady: $\frac{D\overline{u_i}}{Dt} = 0$
- $\tau_{ij} = \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\frac{\partial u_k}{\partial x_k}\delta_{ij}\right) = 0$ (29)

This reduces the momentum equation ((30), 21) to its static form:

$$0 = -\frac{\partial\overline{p}}{\partial x_i} + \overline{\rho g_i} \tag{44}$$

Integrating equation 44 results in the hydrostatic equation. Further needed for the caluclation of the reference state is the equation for the adiabatic temperature gradient (since the reference state temperature is assumed adiabatic)

$$\left(\frac{\partial T}{\partial x_i}\right)_s = \frac{\alpha T g_i}{c_p} \tag{45}$$

And a specification of the reference state equation of state $\overline{\rho} = \overline{\rho}(\overline{T},\overline{\rho})$.

5.2 Nondimensionalization

Now, the linearized equation of state (42) will be nondimensionalized with scaling factors consisting of representative/characteristic values presented in table 5. With these, dimensionless variables (denoted with an asterisk *) are introduced by dividing the dimensional variables with their characteristic counterpart.

Our starting point is Eq. (42), together with the thermodynamical definitions of the thermal expansivity and isothermal compressibility, respectively:

$$\alpha = -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T}\right)_p \tag{46}$$

$$\chi_T = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial p}\right)_T \tag{47}$$

Eq. (42) becomes

$$\begin{aligned} \rho &= \overline{\rho}\,(\overline{T}\,,\overline{p}\,) + \overline{\rho}\,\overline{\chi}_T p' - \overline{\rho}\,\overline{\alpha}\,T' \\ \frac{\rho}{\overline{\rho}} &= 1 + \overline{\chi}_T p' - \overline{\alpha}\,T' \end{aligned} \tag{48}$$

Now, we introduce scaling factors to nondimensionalize the above equation.

- Trivial are the characteristic values χ_{T_r}, α_r, ΔT_r and ρ_r.
- for the velocity scale u_r we introduce the Péclet number, which measures the magnitude of advection relative to diffusion, and assume $\mathrm{Pe} = 1$ (advection and diffusion of same order of magnitude):

$$\begin{aligned} \mathrm{Pe} &= \frac{u_r b}{\alpha_r} = \frac{u_r \rho_r c_{p_r} b}{k_r} = 1 \\ u_r &= \frac{k_r}{\rho_r c_{p_r} b} \end{aligned} \tag{49}$$

- Assuming comparability in magnitude of pressure and viscous forces in the Navier-Stokes equation (with constant dynamic viscosity μ) (Eq. 31):

$$\begin{aligned} \mathcal{O}\left(\frac{\partial p}{\partial x_i}\right) &= \mathcal{O}(\mu \frac{\partial^2 u_i}{\partial x_j^2}) \\ \frac{p_r}{b} &= \mu_r \frac{u_r}{b^2} \\ p_r &= \mu_r \frac{u_r}{b} = \frac{\mu_r k_r}{\rho_r c_{p_r} b^2} \end{aligned} \tag{50}$$

Plugging in $\rho = \rho^* \rho_r$ and similar for the other variables into Eq. (48) gives

$$\begin{aligned} \frac{\rho^*}{\overline{\rho}^*} &= 1 + \overline{\chi}^*_T \chi_{T_r} \frac{p'^* \mu_r k_r}{\rho_r c_{p_r} b^2} - \overline{\alpha}^* \alpha_r T'^* \Delta T_r \\ &= 1 + \overline{\chi}^*_T p'^* \left(\frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2} \cdot \frac{\mu_r C_{p_r}}{k_r}\right) - \overline{\alpha}^* T'^* (\alpha_r \Delta T_r) \\ &= 1 + \overline{\chi}^*_T p'^* \mathrm{M}^2 \mathrm{Pr} - \overline{\alpha}^* T'^* \epsilon \end{aligned} \tag{51}$$

where

$$\mathrm{M}^2 = \frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2}, \quad \mathrm{Pr} = \frac{\mu_r C_{p_r}}{k_r}, \quad \epsilon = \alpha_r \Delta T_r \tag{52}$$

which are the Mach number (squared, measuring the influence of compressibility), Prandtl number (ratio of momentum diffusivity to thermal diffusivity, or, equivalently the ratio between advection and conduction) and the fractional density change due to temperature variations, respectively. I refer to table 6 for an overview of all dimensionless parameters and characteristic values used in the preceding equations.

Plugging in the values from table 5 gives $Pr \approx 2.5 \times 10^{23}$, which is considered practically infinite, and $M^2 \approx 10^{-33}$ and consequently $M^2 Pr \approx 2 \times 10^{-10} \ll 1$. If $\epsilon \ll 1$ as well, $\frac{\overline{\rho}^*}{\overline{\rho}^*} \approx 1$ and density perturbations tue to mantle convection are small compared to the reference hydrostatic density of the mantle.
The dimensionless form of the **conservation of mass equation** (Eq. (18)) can be obtained in the same manner: do I have to do this whole derivation as well?

$$\begin{aligned} &\overline{\rho}^* \, \overline{\chi}_T^* \, M^2 Pr \, \frac{\partial p'^*}{\partial t^*} - \overline{\alpha}^* \, \overline{\rho}^* \, \epsilon \, \frac{\partial T'^*}{\partial t^*} \\ &+ \frac{\partial}{\partial x_i^*} \left[\left\{ \overline{\rho}^* + \overline{\rho}^* \, \overline{\chi}_T^* \, p'^* M^2 Pr - \overline{\rho}^* \, \overline{\alpha}^* \, \epsilon \, T'^* \right\} u_i^* \right] = 0 \end{aligned} \tag{53}$$

The dimensionless and scaled **Navier-Stokes equation** (from Eq. 30 and Eq. 51) is

$$\begin{aligned} &\frac{1}{Pr} \left\{ \overline{\rho}^* + \overline{\rho}^* \, \overline{\chi}_T^* \, M^2 Pr \, p'^* - \overline{\rho}^* \, \overline{\alpha}^* \, \epsilon \, T'^* \right\} \frac{D u_i^*}{D t^*} = - \frac{\partial p'^*}{\partial x_i^*} + \overline{g}_i^* \, \overline{\rho}^* \, \overline{\chi}_T^* \, p'^* \, \frac{D}{\gamma_r} \, \frac{c_{p,r}}{c_{v,r}} \\ &- \overline{g}_i^* \, \overline{\rho}^* \, \overline{\alpha}^* \, T'^* \, Ra + \frac{\partial}{\partial x_j^*} \left\{ \mu^* \left(\frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*} - \frac{2}{3} \, \delta_{ij} \, \frac{\partial u_k^*}{\partial x_k^*} \right) \right\} \end{aligned} \tag{54}$$

where the dimensionless parameter Ra is the Rayleigh number:

$$Ra \equiv \frac{\alpha_r \, \Delta T_r \, \rho_r^2 \, g_r \, b^3 \, c_{p,r}}{\mu_r \, k_r} \tag{55}$$

The Rayleigh number controls the strenght of the buoyancy force term associated with temperature variations, which is the major driving force of thermal convection.

The scaled and nondimensional **energy equation** (41) is

$$\begin{aligned} &\left\{ \overline{\rho}^* + \overline{\rho}^* \, \overline{\chi}_T^* \, M^2 Pr \, p'^* - \overline{\rho}^* \, \overline{\alpha}^* \, \epsilon \, T'^* \right\} \left(\overline{c}_p^* \right) \frac{D}{D t^*} \left(\overline{T}^* + T'^* \right) \\ &\quad - \overline{\alpha}^* \left(\overline{T}^* + T'^* \right) \frac{D p'^*}{D t^*} \, \frac{\epsilon D}{Ra} - \overline{\alpha}^* \left(\overline{T}^* + T'^* \right) \overline{g}_i^* \, u_i^* \, \overline{\rho}^* \, D \\ &= \frac{\partial}{\partial x_i^*} \left\{ \overline{k}^* \, \frac{\partial}{\partial x_i^*} \left(\overline{T}^* + T'^* \right) \right\} + \Phi^* \, \frac{D5}{Ra} \\ &\quad + \left(\overline{\rho}^* + \overline{\rho}^* \, \overline{\chi}_T^* \, M^2 Pr \, p'^* - \overline{\rho}^* \, \overline{\alpha}^* \, \epsilon^* \, T'^* \right) H^* \left(\frac{b^2 H_r \rho_r}{k_r \, \Delta T_r} \right) \end{aligned} \tag{56}$$

5.3 Approximations

In the limit of $\epsilon \to 0$ and $M^2 Pr \to 0$, eq. 53 reduces to

$$\frac{\partial}{\partial x_i^*} \left(\overline{\rho}^* \, u_i^* \right) = 0 \tag{57}$$

Which is known as the **anelastic conservation of mass equation** or the **liquid anelastic approximation** (Jarvis and Mckenzie, 1980). The absence of the $\frac{\partial \rho'}{\partial t}$ term eliminates seismic waves from the set of equations governing mantle convection. This makes sense physically, since seismic activity and mantle convection occur on very different time scales. If, additionally, $\overline{\rho}$ is considered constant with depth, eq. 57 reduces to the incompressible continuity equation:

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{58}$$

This approximation is called the **Boussinesq approximation**.For this approximation to be appropriate, the change in $\overline{\rho}$ with depth has to be investigated. Using the adiabatic compressibility

$$\chi_a = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial p}\right)_s \tag{59}$$

and the hydrostatic equation

$$\frac{\partial \overline{\rho}}{\partial x_i} = \overline{\rho}\,\overline{g_i} \tag{60}$$

gives the density gradient

$$\begin{aligned}\frac{\partial \overline{\rho}}{\partial x_i} &= \left(\frac{\partial \overline{\rho}}{\partial \overline{p}}\right)_s \frac{\partial \overline{p}}{\partial x_i} = \overline{\rho}\,\overline{\chi}_a \frac{\partial P}{\partial x_i} \\ &= \overline{\rho}^{\,2}\chi_a \overline{g}_i\end{aligned} \tag{61}$$

The local density scale height is defined as

$$\begin{aligned}\overline{h}_d &\equiv \left\{\frac{1}{\overline{\rho}}\left|\frac{\partial \overline{\rho}}{\partial x_i}\right|\right\}^{-1} \\ &= (\overline{\rho}\,\overline{\chi}_a \overline{g})^{-1} \\ &= \frac{\overline{\gamma}\,\overline{c}_p}{\overline{\alpha}\,\overline{g}}\end{aligned} \tag{62}$$

where $\gamma = \frac{\alpha}{\rho c_p \chi_a}$ is the Grünseisen parameter. The change in $\overline{\rho}$ with depth is negligible (and the Boussinesq/ incompressible approximation applicable) if $\frac{b}{\overline{h}_d} \ll 1$. Assuming characteristic values, this condition gives

$$\begin{aligned}\frac{b}{\overline{h}_d} &= \frac{1}{\gamma_r}\frac{\alpha_r g_r b}{c_{p_r}} \\ &= \frac{D}{\gamma_r} \ll 1\end{aligned} \tag{63}$$

With D the dissipation number. For the Earth's mantle, $D/\gamma_r \approx 0.5$, and the incompressible mass conservation equation is not very accurate. However, due to its simplicity, it is often adopted into models of mantle convection where the effect of compressibility are minor.

Returning to the (dimensionless and scaled) Navier-Stokes equations, we can see that when

$$\frac{1}{Pr} \to 0 \quad and \quad M^2 Pr \to 0 \tag{64}$$

eq. 54 reduces to its **anelastic (liquid) approximation**:

$$\begin{aligned}0 = &-\frac{\partial p'^*}{\partial x_i^*} + g_i^*\,\overline{\rho}^*\,\overline{\chi}_T^*\,p'^*\,\frac{D}{\gamma_r}\,\frac{c_{p,r}}{c_{v,r}} - g_i^*\,\overline{\rho}^*\,\overline{\alpha}^*\,T'^*\,Ra \\ &+ \frac{\partial}{\partial x_j^*}\left\{\mu^*\left(\frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*} - \frac{2}{3}\,\delta_{ij}\,\frac{\partial u_k^*}{\partial x_k^*}\right)\right\}\end{aligned} \tag{65}$$

and the dimensional form

$$0 = -\frac{\partial p'}{\partial x_i} + \overline{\rho}\,\overline{g}_i\,(\overline{\chi}_T\,p' - \overline{\alpha}\,T') + \frac{\partial}{\partial x_j}\left\{\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\,\delta_{ij}\,\frac{\partial u_k}{\partial x_k}\right)\right\} \tag{66}$$

With the additional limit of $D/\gamma_r \to 0$ (together with incompressibility and $\overline{\rho^*} = \overline{\alpha^*} = 1$) the dimensionless Navier-Stokes equations further reduce to the **Boussinesq approximation**:

$$0 = -\frac{\partial p'^*}{\partial x_i^*} - \overline{g}_i^*\,T'^*\,Ra + \frac{\partial}{\partial x_j^*}\left\{\mu^*\left(\frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*}\right)\right\} \tag{67}$$

and the dimensional form

$$0 = -\frac{\partial p'}{\partial x_i} - \overline{\rho}\,\overline{g}_i\,\overline{\alpha}\,T' + \frac{\partial}{\partial x_j}\left\{\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right\} \tag{68}$$

Returning to the (nondimensional) heat equation (56) and applying the conditions of $M^2 Pr \to 0$ and $\epsilon \to 0$ for the **anelastic approximation**, this reduces to

$$\begin{aligned}&\overline{\rho}^{\,*}\,\overline{c}_p^{\,*}\,\frac{D}{Dt^*}\left(\overline{T}^{\,*} + T'^{*}\right) - \overline{\alpha}^{\,*}\left(\overline{T}^{\,*} + T'^{*}\right)\overline{g}_i^{\,*}\,u_i^*\,\overline{\rho}^{\,*}\,D \\ &= \frac{\partial}{\partial x_i^*}\left\{\overline{k}^{\,*}\,\frac{\partial}{\partial x_i^*}\left(\overline{T} + T'\right)\right\} + \Phi^*\,\frac{D}{Ra} + \overline{\rho}^{\,*}\,H^*\left(\frac{b^2 H_r \rho_r}{k_r\,\Delta T_r}\right)\end{aligned} \tag{69}$$

Now, using thermodynamic relations in combination with the hydrostatic equation (6.9.17+6.9.18) give

$$\left(\frac{\partial T}{\partial x_i}\right)_s = \frac{\alpha T g_i}{c_p} \tag{70}$$

or, in dimensionless form

$$\left(\frac{\partial \overline{T}^{\,*}}{\partial x_i^*}\right)_s = \frac{\overline{\rho}^{\,*}\overline{T}^{\,*} D \overline{g}_i^{\,*}}{\overline{c}_p^{\,*}} \tag{71}$$

Where the dissipation number D arises from the nondimensionalization. With this relation, the $\overline{T}^{\,*}$ terms in the RHS of equation 69 cancel out and the **anelastic, dimensionless form of the energy equation** becomes

$$\begin{aligned}\overline{\rho}^{\,*}\,\overline{c}_p^{\,*}\,\frac{DT'^{*}}{Dt^*} - T'^{*}\,\overline{\alpha}^{\,*}\,\overline{g}_i^{\,*}\,u_i^*\,\overline{\rho}^{\,*}\,D = \frac{\partial}{\partial x_i^*}\left\{\overline{k}^{\,*}\,\frac{\partial}{\partial x_i^*}\left(\overline{T}^{\,*} + T'^{*}\right)\right\} \\ + \Phi^*\,\frac{D}{Ra} + \overline{\rho}^{\,*}\,H^*\left(\frac{b^2 H_r \rho_r}{k_r\,\Delta T_r}\right)\end{aligned} \tag{72}$$

and in dimensional form

$$\overline{\rho}\,\overline{c}_p\,\frac{DT'}{Dt} - T'\,\overline{\alpha}\,\overline{g}_i\,u_i\,\overline{\rho} = \frac{\partial}{\partial x_i}\left\{\overline{k}\,\frac{\partial}{\partial x_i}\left(\overline{T} + T'\right)\right\} + \Phi + \overline{\rho}\,H \tag{73}$$

If $\overline{k}^{\,*}$, $\overline{\alpha}^{\,*}$, $\overline{c}_p^{\,*}$ and $\overline{g}_i^{\,*}$ are considered constant, then the terms representing heat conduction along the adiabat become INSERT EQ 6.10.31-32 HERE where eq. 71 was used.

Assuming $D \to 0$, which physically means assuming the representative thermal expansivity α_r to be zero, we arrive at the **Boussinesq energy equation**. INSERT EQ. 6.10.34-48 HERE

6 Appendix 1: tables of constants and reference/scaling values

7 Jarvis and Mckenzie, 1980 General equations

Conservation of mass, momentum and energy:

ADD ALGEBRA COMPARISON SCHUBERT in final doc

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0 \tag{74}$$

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u}\right) = -\nabla P + \rho g + \frac{\partial}{\partial x_j}\tau_{ij} \tag{75}$$

$$\rho C_p\left\{\frac{\partial T}{\partial t} - \frac{\alpha T}{\rho C_p}\frac{\partial P}{\partial t} + \boldsymbol{u} \cdot [\nabla T - (\nabla T)_s]\right\} = \nabla \cdot (K \nabla T) + H + \tau_{ij}\frac{\partial u_i}{\partial u_{ij}} \tag{76}$$

All terms and symbols used in these equations are shown in table 8. The (Newtonian) relation between the deviatoric stress tensor τ_{ij} and the velocity field is

Table 4: Constants used by Schubert et al., 2001

Constant	Value	Name
k_b	0	Bulk viscosity
α		thermal expansivity
χ_T		isothermal compressibility
c_v		specific heat at constant volume
c_p		specific heat at constant pressure
χ_a		adiabatic compressibility
k		thermal conductivity
H		rate of internal heat production per unit mass
K_a		adiabatic bulk modulus
K_T		isothermal bulk modulus
Φ	$\tau_{ij}\frac{\partial u_i}{\partial x_j}$	dissipation function (eq. 6.9.3)

Table 5: Reference values used by Schubert et al., 2001

Symbol	Value	Description
ΔT_r		Characteristic temperature difference driving thermal convection
χ_{T_r}		Representative isothermal compressibility
α_r		Representative thermal expansivity
u_r	$\frac{k_r}{\rho_r c_{p_r} b}$	Velocity scale (follows from assuming $Pe = 1$)
μ_r	10^{21} Pa s	Representative viscosity
b		Depth of convecting region
p_r	$\frac{\mu_r u_r}{b}$	Scale for dynamic pressure variations
k_r	4 W m^{-1} K^{-1}	Reference value thermal conductivity
c_{p_r}	1 kJ kg^{-1} K^{-1}	Reference specific heat at constant pressure
t_r	$\frac{b}{u_r} = \frac{\rho_r c_{p_r} b^2}{k}$	reference timescale

$$\tau_{ij} = \eta\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\,\delta_{ij}\,\frac{\partial u_k}{\partial x_k}\right) + \eta_2\,\delta_{ij}\,\frac{\partial u_k}{\partial x_k}. \tag{77}$$

Combining Eq. (77) with the momentum equation (75) gives the general Navier-Stokes equations, but Jarvis and Mckenzie (1980) does not show these. Instead, they assume that $\frac{1}{Pr} \to 0$, which they assume (do they?) is the case for the Earth's mantle, the inertial forces can be ignored. The resulting equations will be discussed in the next section, which covers the approximations made by Jarvis and Mckenzie (1980).

8 Jarvis and Mckenzie, 1980 Approximations

Problem statement: for the Boussinesq approximation to be valid, the depth range of convection d must be small compared to the temperature scale height H_T ($d/H_T \ll 1$). This is not the case for many systems of deep convection, amongst which the convection of planetary mantles. Physically, $d/H_T \approx O(1)$ means that the influence of viscous dissipation, variable adiabatic temperature- and density gradients must be maintained in the set of governing equations.
The assumptions made for the **anelastic liquid approximation** are that $d = O(H_T)$ and that $\alpha T \ll 1$

Table 6: Dimensionless parameters and their values for the Earth's mantle

Parameter	Definition	Value	Description
Pr	$\frac{\mu_r c_{p_r}}{k_r}$	2.5×10^{23}	Prandtl number
M^2	$\frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2}$	10^{-33}	Mach number (squared)
ϵ	$\alpha_r \Delta T_r$		Fractional density change due to temperature variations
Pe	$\frac{u_r b}{\alpha_r} = \frac{u_r \rho_r c_{p_r} b}{k_r}$		Péclet number for heat transfer (advection/diffusion)

what does this mean and how does this relate to Prandtl number? and if $\alpha \ll 1$, how come D is not $\ll 1$?

8.1 Reference state and equation of state

A reference state with hydrostatic pressure P_H and adiabatic temperature distribution (gradient?) T_s is considered. The equation of state is written as a linear Taylor expansion of ρ about the reference state density distribution ρ_r:

$$\rho(T, P) = \rho_r[1 - \alpha(T - T_s) + K_T^{-1}(P - P_H)] \tag{78}$$

Where

$$\alpha = \frac{1}{\rho_r}\left(\frac{\partial \rho_R}{\partial T}\right)_P \tag{79}$$

$$K_T = \rho_r \left(\frac{\partial P}{\partial \rho_r}\right)_T \tag{80}$$

Determining the reference density distribution ρ_r is done by considering the gradient of $\rho_r(S, P_H)$ with S the entropy per unit mass:

$$\nabla \rho_R(S, P_H) = \left(\frac{\partial \rho_r}{\partial S}\right)_{P_H} \nabla S + \left(\frac{\partial \rho_r}{\partial P_H}\right)_S \nabla P_H \tag{81}$$

With $\nabla S = 0$ since the reference state is adiabatic, and with $\nabla P_H = -\rho_r g \hat{\boldsymbol{z}}$:

$$\frac{1}{\rho_r}\frac{\partial \rho_r}{\partial z} = -\frac{\rho_r g}{K_s} \tag{82}$$

with $K_s = \rho_r \left(\frac{\partial P_H}{\partial \rho_r}\right)_s$. Using the definition of the Grüneisen parameter Γ and the temperature scale height of the liquid H_T gives the Adams-Williamson relation add images!

$$\frac{1}{\rho_r}\frac{\partial \rho_r}{\partial z} = -\frac{1}{H_T \Gamma} \tag{83}$$

Which is valid for a **chemically homogeneous adiabatic density distribution under hydrostatic pressure**. Integration of equation 83 over z gives the reference densitiy distribution as

$$\rho_r(z) = \rho_0 \exp[(d - z)/H_T \Gamma] \tag{84}$$

where $\rho_0 = \rho_r(z = d)$ is the reference density value at the surface orientation of coordinate system?? Plugging this expression of the reference density distribution into 78 gives the (linearized equation) of state

$$\rho(T, P) = \rho_0 \exp[(d - z)/H_T \Gamma][1 - \alpha(T - T_s) + K_T^{-1}(P - P_H)] \tag{85}$$

8.2 Nondimensionalization

9, 9. move tables here? In this section, Jarvis and Mckenzie, 1980 introduces various characteristic values of variables in order to nondimensionalize the preceding equations. These characteristic values and the resulting dimensionless variables are presented in the tables below.The characteristic value for **velocity** is taken directly from McKenzie et al., 1974, and their derivation is expanded below:

Our starting point is the momentum equation (in the notation of McKenzie et al., 1974), with the left-hand side set to 0 (on the basis of $\mathrm{Re} \ll 1$ and $\frac{1}{\mathrm{Pr}} \to 0$):

$$0 = -\frac{1}{\rho}\frac{\partial P}{\partial x_i} + \frac{\partial \phi}{\partial x_i} + \frac{1}{\rho}\frac{\partial \tau_{ij}}{\partial x_j} \tag{86}$$

with

$$\tau_{ij} = \rho\nu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$
$$\nabla\phi = \boldsymbol{g}$$

last term in Eq. (86) becomes help index confusion

$$\begin{aligned}\frac{1}{\rho}\frac{\partial \tau_{ij}}{\partial x_j} &= \rho\nu\left(\frac{\partial}{\partial x_j}\frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j}\frac{\partial u_j}{\partial x_i}\right)\\ &= \rho\nu(\nabla^2 u_i + \frac{\partial}{\partial x_i}\nabla\cdot\boldsymbol{u})\end{aligned}$$

assuming incompressibility, $\nabla\cdot\boldsymbol{u} = 0$, and or should this be $\boldsymbol{u}$ (vector)?

$$\frac{1}{\rho}\frac{\partial \tau_{ij}}{\partial x_j} = \nu\nabla^2 u_i \tag{87}$$

Now we formulate an equation of state under the Boussinesq approximation (Taylor expansion of density $\rho(T)$ around a reference state temperature T_0):

$$\begin{aligned}\rho &= \rho_0[1-\alpha(T-T_0)]\\ \frac{1}{\rho} &= \rho_0^{-1}[1-\alpha(T-T_0)]^{-1}\\ &\approx \rho_0^{-1}[1+\alpha(T-T_0)]\end{aligned}$$

with $\alpha = -\frac{1}{\rho}\left(\frac{\partial\rho}{\partial T}\right)_P$ the thermal expansivity coefficient.

We can split the pressure gradient ∇P into a hydrostatic part (∇P_s) and a non-hydrostatic part ∇p:

$$\begin{aligned}\nabla P &= \nabla P_s + \nabla p\\ &= \rho_0\boldsymbol{g} + \nabla p\end{aligned}$$

Together with the equation of state, the pressure term + gravity in Eq. (86) then becomes

$$\begin{aligned}-\frac{1}{\rho}\nabla P + \boldsymbol{g} &= -\rho_0^{-1}(1+\alpha(T-T_0))(\nabla P_s + \nabla p) + \boldsymbol{g}\\ &= -\rho_0^{-1}\nabla P_s - \rho_0^{-1}\nabla P_s\alpha(T-T_0) - \rho_0^{-1}\nabla p - \rho_0^{-1}\nabla p\alpha(T-T_0) + \boldsymbol{g}\\ &\approx -\boldsymbol{g}\alpha(T-T_0) - \rho_0^{-1}\nabla p\end{aligned}$$

Where we neglected the second-order terms ($\sim \nabla pT$) and used that $\nabla P_s = \rho_0 \boldsymbol{g}$, cancelling out the gravity terms. With this, Eq. (86) becomes

$$\nu \nabla^2 \boldsymbol{u} = \boldsymbol{g}\alpha(T - T_0) + \rho_0^{-1} \nabla p \tag{88}$$

note that ∇p is only the non-hydrostatic pressure gradient.

We can take the curl of Eq. (88), using that $\nabla \times \boldsymbol{u} = \boldsymbol{\omega}$, that the curl of a gradient is always zero ($\nabla \times (\nabla f) = 0$), and that $\boldsymbol{g} = \nabla \phi$:

$$\begin{aligned} \nu \nabla^2 \boldsymbol{\omega} &= \nabla \times (\boldsymbol{g}\alpha(T - T_0)) + \nabla \times (\rho_0^{-1} \nabla p) \\ &= \nabla \times [\boldsymbol{g}\alpha(T - T_0)] \\ &= \alpha(T - T_0)(\nabla \times \boldsymbol{g}) + \alpha \nabla T \times \boldsymbol{g} \\ &= \alpha \nabla T \times \boldsymbol{g} \end{aligned} \tag{89}$$

Now we assume that the flow in restricted to the x, z plane, meaning $\boldsymbol{u} = (u, 0, w) = \left(-\frac{\partial \psi}{\partial z}, 0, \frac{\partial \psi}{\partial x}\right)$ (where ψ is the stream function) and thus $\boldsymbol{\omega} = (0, \omega, 0) = (0, -\nabla^2 \psi, 0)$. With $\boldsymbol{g} = (0, 0, g)$, Eq. 89 becomes (computing the cross product)

$$\nabla^2 \omega = \frac{\alpha g}{\nu} \frac{\partial T}{\partial x} \tag{90}$$

From this equation, we can get an characteristic velocity for the convecting system.

$$\begin{aligned} \mathcal{O}(\nabla^2 \omega) &= \mathcal{O}(\frac{\alpha g}{\nu} \frac{\partial T}{\partial x}) \\ \frac{U}{d^3} &= \frac{\alpha g}{\nu_0} \frac{T_1}{d} \\ U &= \frac{\alpha g T_1 d^2}{\nu_0} \end{aligned}$$

Where U, ν_0, and T_1 are characteristic values for velocity, kinematic viscosity, and temperature deviations from the adiabatic reference profile T_0, respectively.

For the scaling of pressure variations P_1 TO DO

Table 7: characteristic values of Jarvis and Mckenzie, 1980

Variable	Characteristic value	Dimensionless variable
depth	d	$(x', z') = \frac{(x,z)}{d}$
temperature	$T_1 (= 1\text{K})$	$T' = \frac{(T - T_0)}{T_1}$
density	ρ_0	$\rho_r' = \frac{\rho_r}{\rho_0}, \quad \rho = \frac{\rho}{\rho_0}$
velocity	$U = \frac{g\alpha T_1 d^2}{\nu_0}$	
kinematic viscosity	$\nu_0 = \eta / \rho_0$	
time	$\tau = d/U$	

8.3 The anelastic-liquid approximation

From now on, only the dimensionless variables will be used but with primes surpressed (i.e. all variables are assumed dimensionless). The dimensionless equation of state (from eq. 85) becomes

$$\rho = \rho_r [1 - \mu(T - T_s) + \mu(\rho_0 g d / K_T) P_1] \tag{91}$$

with $\mu = \alpha T_1$. If $\mu \to 0$, ρ reduces to its reference profile ρ_r (as is the case in the Boussinesq approximation).

9 Appendix 1: tables of symbols

Table 8: Symbols used by Jarvis and Mckenzie, 1980

d		depth of the convecting layer
H_T	$\frac{C_p}{g\alpha}$	temperature scale height of the liquid
C_p		specific heat at constant pressure
C_v		specific heat at constant volume
α		volume coefficient of thermal expansion
τ_{ij}		deviatoric stress tensor
$(\nabla T)_s$	$\frac{\alpha T}{\rho C_p}\nabla P$	adiabatic temperature gradient
K		thermal conductivity
H		rate of internal heat production
η	$\rho\nu$	dynamic viscosity
ν		kinematic viscosity
η_0	$\ll 1$	second or bulk viscosity
Γ	$\frac{\alpha K_s}{\rho C_p} = \frac{\alpha K_T}{\rho C_v}$	Grüneisen's parameter, assumed constant and $O(1)$
K_s		isentropic modulus of bulk compressibility
K_T		isothermal modulus of bulk compressibility
μ	αT_1	
α	$10^{-5} \quad {}^\circ K^{-1}$	volume coefficient of thermal expansion

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Table 9: characteristic values of Jarvis and Mckenzie, 1980

depth	d
temperature	$T_1 = \frac{1}{\mathrm{K}} WHAT?? \to 1°\mathrm{K}$
density	ρ_0
velocity	$U = \frac{g\alpha T_1 d^2}{\nu_0}, \quad \nu_0 = \eta/\rho_0$help this is different from schubert
time	$\tau = d/U$

Table 10: Dimensionless variables in Jarvis and Mckenzie, 1980

(x', z')	$(x, z)/d$
ρ'_r	$\rho_r/\rho_0 = \exp[(1 - z')D/\Gamma], \quad D = d/H_T$
ρ'	ρ/ρ_0
T'	$(T - T_0)/T_1$
t'	t/τ
P'_1	$\frac{P_1 d}{U\eta} = P_1 \alpha T_1 \rho_0 g d, \quad P_1 = P - P_H$
κ'_0	$\kappa_0\nu_0/g\alpha T_1 d^3$, thermal diffusivity
ϵ'_0	$\epsilon_0\nu_0/g\alpha T_1^2 d$, energy generation rate, $\epsilon_0 = H/\rho_0 C_p$
λ	L/d, box width, L is the dimensional box width